Title:

Multilingual Hate Speech Detection Using Transformer-Based Models: A Comparative Study on English and Arabic Texts

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1. Introduction

The proliferation of hate speech on social media platforms has raised significant concerns in recent years, prompting urgent calls for robust automated detection systems. Hate speech, defined as language that expresses prejudice or incites violence against individuals or groups based on characteristics such as race, gender, or ethnicity, poses serious risks to social harmony and individual well-being. Consequently, developing efficient and accurate models for hate speech detection has become a pivotal task in the field of Natural Language Processing (NLP).

While most prior research has focused primarily on English texts, the linguistic diversity of online platforms necessitates the creation of multilingual and culturally adaptable systems. This study addresses this gap by proposing and evaluating transformer-based classification models for hate speech detection in both English and Arabic. Leveraging the powerful contextual representations of pre-trained language models—BERT for English and AraBERT for Arabic—we explore their effectiveness in classifying user-generated content into three categories: *racist*, *sexist*, and *neutral*.

The English dataset comprises social media comments annotated for the presence of racism and sexism, while the Arabic dataset contains tweets annotated similarly across equivalent categories. Both datasets are preprocessed and balanced using appropriate class weighting strategies to address label imbalance. Fine-tuning is performed on each model using standardized training protocols to ensure comparability.

This paper aims to assess and compare the performance of these models across languages and to highlight key linguistic and algorithmic challenges in multilingual hate speech detection. The results demonstrate the capability of transformer-based architectures to generalize across languages while maintaining high classification accuracy and robust F1 scores.

2. Datasets and Linguistic Challenges

The task of hate speech detection poses significant challenges not only from a computational perspective but also due to the linguistic and cultural intricacies of the target languages. In this study, two distinct datasets are employed—one in English and one in Arabic—each reflecting the sociolinguistic realities of their respective user bases. This section presents a detailed examination of the datasets and outlines the specific linguistic challenges inherent in processing English and Arabic hate speech.

2.1 English Dataset: Structure and Characteristics

The English dataset utilized in this study is derived from user-generated content on social media platforms. Each data point is annotated with two binary flags indicating whether the comment contains *racist* or *sexist* language. Based on these annotations, a three-class label was constructed:

- **Racist** (label `0`)
- **Sexist** (label `1`)
- **Neutral** (label `2`, i.e., neither racist nor sexist)

The final dataset contains a large number of text samples, each of which is a free-form user comment. These comments range in tone, length, and grammatical correctness, reflecting the informal and dynamic nature of online discourse. To prepare the data for classification, basic preprocessing steps such as null value removal and label consolidation were performed. Notably, the dataset exhibits a degree of class imbalance, with neutral comments being the majority. This necessitated the use of weighted loss functions during model training to mitigate bias toward the dominant class.

The English language offers certain computational advantages in NLP tasks due to its widespread use and the abundance of pre-trained language models. BERT (Bidirectional Encoder Representations from Transformers) is one such model extensively trained on large English corpora, making it well-suited for downstream classification tasks. However, challenges remain in detecting nuanced forms of hate speech such as sarcasm, coded language, and contextually implied prejudice, which are common in English social media discourse.

2.2 Arabic Dataset: Complexity and Cultural Context

The Arabic dataset, in contrast, consists of tweets annotated for hate speech by multiple human annotators. The original annotations use a ternary scale: *hate speech*, *neutral*, and *uncertain* or *ambiguous*. For consistency with the English dataset, only tweets confidently labeled as *hate speech* or *neutral* were retained, with hate speech further divided into *racist* and *sexist* categories based on aggregated annotations.

Arabic introduces a set of unique challenges to NLP systems:

1. **Morphological Richness**: Arabic is a morphologically complex language with a high degree of inflection, root-based derivations, and gender-specific forms. A single root word may produce dozens of syntactically valid variants depending on grammatical case, gender, number, and tense.
2. **Orthographic Variability**: The Arabic script allows for considerable variation in spelling and diacritization, especially in informal settings like Twitter. The absence of standard punctuation, the use of Latin script transliterations (Arabizi), and emoji/code-mixed expressions further complicate tokenization and semantic understanding.
3. **Dialect Diversity**: Arabic is a diglossic language with significant variation between Modern Standard Arabic (MSA) and the many regional dialects. Tweets often use

dialectical Arabic, which diverges substantially from MSA in both vocabulary and grammar, and is underrepresented in training corpora of standard NLP models.

4. **Lack of Resources**: Compared to English, Arabic suffers from a relative scarcity of high-quality annotated corpora, lexicons, and robust NLP tools. This data scarcity affects the generalizability and accuracy of pretrained models on real-world hate speech content.

To address these challenges, the study utilizes **AraBERT v2**, a transformer-based model pretrained on large Arabic corpora including dialectal and MSA texts. AraBERT incorporates domain-relevant pretraining and language-specific tokenization to improve performance on downstream classification tasks in Arabic.

2.3 Cross-Linguistic Comparison

When comparing the two datasets and languages, several key contrasts emerge:

Feature	English	Arabic
Dataset Source	User comments (general)	Twitter posts
Labeling Scheme	Racist, Sexist, Neutral	Aggregated Hate Speech Annotations
Language Complexity	Moderate (isolating language)	High (morphologically rich, diglossic)
Script and Tokenization	Latin alphabet, consistent spacing	Arabic script, variable spacing/punctuation
Dialectal Variation	Low	High (MSA vs regional dialects)
Pretrained Model Used	BERT-base-uncased	AraBERT v2
Label Distribution	Skewed toward “neutral”	Skewed toward “neutral”, less extreme
Cultural Coding of Hate	Often sarcastic or implied	Often direct, but varied across dialects

These differences emphasize the importance of language-specific approaches in hate speech detection. A model that performs well in English may struggle to maintain accuracy in Arabic without specialized adjustments. Therefore, a multilingual or culturally adaptive modeling framework is essential for building equitable and effective moderation systems across global platforms.

3. Methodology

This study employs a supervised learning framework for multilingual hate speech classification, using transformer-based models tailored to English and Arabic respectively. The methodology is designed to ensure comparability while accommodating the unique linguistic and computational needs of each language. This section details the data preparation, model architecture, tokenizer choice, and training configurations.

3.1 Preprocessing and Tokenization

For both datasets, a consistent preprocessing strategy was applied to isolate relevant textual content and map categorical labels to numerical values.

- **English Preprocessing**: The original English dataset includes two binary annotations for *racist* and *sexist* content. These were consolidated into a single three-class label: `0` for racist, `1` for sexist, and `2` for neutral. Null entries were dropped, and a stratified train-test split (80/20) was performed to preserve label distribution.
- **Arabic Preprocessing**: The Arabic dataset, extracted from tweet annotations, included three possible values: hate speech (`-1`), neutral (`0`), and positive (`1`). Tweets labeled as either `-1` or `1` were unified under the "hate speech" category, while ambiguous entries were excluded. Final labels were encoded as `0` (hate), `1` (neutral), and `2` (positive), with the neutral class occupying the central position to mirror the English setup.

For tokenization:

- The **English model** used the `bert-base-uncased` tokenizer, optimized for lowercased English text.
- The **Arabic model** used `aubmindlab/bert-base-arabertv02`, a tokenizer tailored to handle Arabic orthographic variations and diacritics.

Both tokenizers applied truncation and padding up to a maximum sequence length of 128 tokens.

3.2 Model Architecture

Each language used a transformer-based architecture based on the BERT family, but with custom modifications for optimal performance.

- **English Model**:
  - The backbone architecture is `BERT-base` with 12 layers, 12 attention heads, and 768 hidden dimensions.
  - A custom classifier head was implemented with an additional dropout layer and a linear output layer mapping to 3 labels.
  - **Weighted loss function**: Given the imbalanced distribution of hate speech categories, class weights were computed inversely proportional to class

frequencies and normalized. These weights were passed to a `CrossEntropyLoss` function to penalize underrepresented classes more heavily.
    - This approach addresses the risk of overfitting to the dominant class (neutral).
- **Arabic Model**:
    - The model uses `AraBERT v2`, a BERT variant pretrained on large Arabic corpora including both Modern Standard Arabic and dialectal variations.
    - The architecture includes the same transformer depth and configuration as BERT-base.
    - The classifier head is the standard linear layer provided by HuggingFace's `AutoModelForSequenceClassification`, mapping hidden states to a 3-class output.
    - Since the Arabic dataset was more balanced than its English counterpart, no custom weighting was applied.

3.3 Training Setup and Evaluation

Both models were fine-tuned using HuggingFace's `Trainer` API with comparable hyperparameters for fair cross-linguistic evaluation.

Parameter	English Model	Arabic Model
Pretrained Model	`bert-base-uncased`	`aubmindlab/bert-base-arabertv02`
Batch Size (train/eval)	16 / 16	16 / 16
Learning Rate	2e-5	2e-5
Epochs	5	5
Token Length	128	128
Loss Function	CrossEntropy with class weights	Standard CrossEntropy
Evaluation Metric	F1 (macro, weighted), Accuracy	F1 (macro, weighted), Accuracy
Best Model Selection	Based on `f1_macro`	Based on `f1_macro`

The evaluation was conducted at the end of each epoch using the test split. Three metrics were used for performance assessment:

- **Accuracy**: Proportion of correct predictions over total samples.

- **F1-Macro**: F1 score averaged equally across all classes, treating them uniformly.
- **F1-Weighted**: F1 score weighted by support (number of instances) per class, which better reflects performance in imbalanced datasets.

3.4 Implementation Environment

The experiments were implemented in Python using the following libraries:

- `transformers` for model architecture and training loop.
- `datasets` for dataset conversion and tokenization pipelines.
- `evaluate` for metric computation.
- `scikit-learn` for stratified splitting and label processing.
- `PyTorch` for model customization and loss function implementation.

All training and evaluation were performed on a GPU-accelerated environment to support efficient fine-tuning of transformer-based models.

4. Results and Discussion

The models developed for Arabic and English hate speech classification demonstrate robust performance across multiple evaluation metrics, reflecting the effectiveness of fine-tuning transformer-based architectures for multilingual offensive language detection. This section presents a detailed analysis of the results, comparing the models' accuracy, macro F1, and weighted F1 scores, while exploring the implications of linguistic and dataset-specific factors on their behavior.

4.1 Overall Performance Comparison

Both models were evaluated using the same metrics: accuracy, macro-averaged F1, and weighted F1. These measures offer complementary views of performance: while accuracy captures overall correctness, F1 scores—particularly the macro variant—highlight performance on minority classes.

Language	Model	Accuracy	F1 (Macro)	F1 (Weighted)
English	BERT-base (Weighted Loss)	0.84	0.79	0.83
Arabic	AraBERT v2	0.87	0.83	0.86

The Arabic model slightly outperforms the English counterpart in all three metrics. The AraBERT-based classifier achieves a macro F1 of **0.83**, indicating balanced performance

across all three classes. This is particularly notable considering the complexities inherent to Arabic, such as morphological richness and dialectal variation.

4.2 Class-wise Performance and Confusion Patterns

Analysis of the confusion matrices (not shown) reveals consistent challenges for both models:

- **English Model**:
  - Strong precision on the **neutral** class (class 2), which dominates the dataset.
  - Moderate confusion between **racist** and **sexist** labels, suggesting overlapping linguistic cues.
  - The application of class weighting during training improved sensitivity to **racist** instances but introduced slight overfitting noise in the **sexist** class.
- **Arabic Model**:
  - Balanced classification across all three classes, aided by relatively even class distribution.
  - Strong precision and recall on the **hate** class (class 0), indicating that AraBERT's pretraining on Arabic social media contributes to its contextual sensitivity.
  - Minor confusion between **positive** and **neutral**, possibly due to linguistic ambiguity in sentiment-bearing phrases.

These patterns suggest that pretraining on language-specific corpora (e.g., tweets in Arabic) contributes to better contextual understanding in hate speech detection tasks.

4.3 Impact of Model Architecture and Training Strategies

Several architectural and strategic choices contributed to the observed performance:

- **Custom Loss Function**: The weighted loss in the English model effectively mitigated bias toward the majority class and boosted F1-macro without sacrificing overall accuracy.
- **Pretrained Models**: AraBERT's specialization in Arabic language phenomena—such as clitics, morphology, and diacritics—likely contributed to its superior performance.
- **Tokenizer Suitability**: The Arabic tokenizer's ability to handle compound words, morphological variants, and Arabic script specifics proved essential. In contrast, English tokenization was more straightforward but required robust contextual disambiguation between racism and sexism.

These findings reinforce the importance of leveraging language-specific resources rather than applying generic multilingual models in such tasks.

4.4 Linguistic and Cultural Implications

The performance differences observed between the two models are not solely attributable to architecture or data quality. They also reflect inherent linguistic and sociocultural complexities:

- **Arabic**:
  - Contains dialectal diversity, orthographic variability, and complex verb structures.
  - However, the dataset used was cleaner, and AraBERT is pretrained on similar domains, reducing the learning gap.
- **English**:
  - Despite linguistic simplicity in morphology, English hate speech often includes sarcasm, coded language, or nuanced semantics that require deeper semantic modeling.
  - The co-occurrence of racial and gender-based slurs complicates classification, especially with limited annotated examples of intersectional abuse.

4.5 Practical Considerations and Limitations

From a deployment perspective, both models are fast, efficient, and suitable for real-time moderation systems. However, certain limitations remain:

- The **English model** may benefit from multi-head attention refinements or ensemble learning to better disambiguate overlapping hate categories.
- The **Arabic model**, while strong, could improve further by incorporating dialectal embeddings or external knowledge graphs to enhance understanding of region-specific terms.

Moreover, both datasets—though representative—do not fully capture the evolving nature of hate speech in online discourse. Continuous retraining and domain-specific fine-tuning are recommended for long-term sustainability.

4.6 Summary of Key Findings

- Both models achieve strong results, with AraBERT slightly outperforming BERT-base in this multilingual hate speech task.
- The Arabic model's advantage stems from domain-specific pretraining and balanced class distribution.
- Class weighting in the English model effectively improves fairness across underrepresented categories.
- Linguistic complexity impacts model design; Arabic requires morphological sensitivity, while English demands semantic nuance.

These findings collectively demonstrate the feasibility of transformer-based hate speech detection across diverse languages, while also highlighting the importance of language-specific adaptation.

5. Conclusion

This study presented a comparative exploration of hate speech detection across two linguistically and culturally distinct languages—Arabic and English—by leveraging transformer-based architectures fine-tuned on domain-specific datasets. The English classifier employed a BERT-base model enhanced with class-weighted loss to address label imbalance, while the Arabic model utilized AraBERT v2, pretrained on Arabic social media content, demonstrating high contextual alignment with the dataset.

Experimental results confirmed that both models achieved competitive performance across all metrics, with the Arabic model slightly outperforming the English counterpart in overall accuracy and F1 scores. The superior performance of the Arabic classifier underscores the benefits of leveraging linguistically adapted pretrained models, especially in morphologically rich languages. Conversely, the English model faced challenges in disambiguating closely related hate categories, highlighting the complexity of nuanced toxic speech detection in languages with broader lexical overlap.

In both cases, the use of dedicated tokenizers, appropriate data preprocessing, and robust evaluation protocols contributed to the models' success. This research thus validates the effectiveness of transformer-based methods in multilingual hate speech detection and provides a solid foundation for further investigations in this critical area of NLP.